



A vibration-insensitive-cavity design holds impact of higher than 100g

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Abstract

In this paper, we report a robust and vibration insensitive optical cavity design. The robustness of the supporting structure was tested by dropping an aluminum and a ULE cavity model mounted in the supporting frame onto an optical table from a height of about 250 and 20 mm repetitively. The impact acceleration on hitting the table was estimated to be higher than 100g. The success of the test demonstrated that the design was very robust. Meanwhile, the vibration sensitivities of the cavity in three orthogonal directions were measured to be $0.8 \times 10^{-10}/g$, $2.5 \times 10^{-10}/g$ and $1.5 \times 10^{-10}/g$, respectively. The robustness and low vibration sensitivity suggested that the design was suitable for transportable system and might be a good candidate for space applications.

Keywords Optical cavity · Ultra-stable laser · Transportable · Space

1 Introduction

Lasers with linewidth of Hz or sub-Hz level and instability of lower than 10^{-14} are usually called ultra-stable lasers [1–4]. They are widely used in precision physics experiments and metrology, e.g., optical atomic clocks [5–9], local Lorentz invariance test [10–12], dark matters searching [13, 14], and gravitational wave detection [15, 16]. In recent years, transportable optical clocks were developed [17, 18]. Optical clocks and gravitational wave detection projects in space were also proposed [19–21]. In these projects, robust ultra-stable lasers were foreseen to be used.

Nowadays, ultra-stable lasers are usually obtained by stabilizing their frequencies to resonances of high-finesse ultra-stable optical cavities. When a laser's frequency is perfectly locked to a cavity's resonance, the frequency stability of the laser is determined by the length stability of the cavity. Therefore, improving the length stability of the reference cavity is one of the major tasks for building an ultra-stable

laser system. The length stability of a cavity is decided by the Brownian noise limit, temperature, and vibration sensitivities. The former two are mainly decided by the material of the cavity, while the latter is decided by the design and fabricate precision. Usually, in the laboratory, the vibration sensitivities of the cavities were minimized by applying certain cut to the spacers and laying on several magic supporting positions [22, 23]. With this method, the vibration sensitivities of the cavities were reduced to the level of $10^{-10}/g$. Combined with vibration isolation systems, the cavity length instabilities caused by vibration were reduced to be lower than 10^{-16} [3, 4]. However, these supporting structure were very fragile and not suitable for transportable applications. For transportable applications, the cavities were squeezed to fix on their supporting frames [24–28]. With the designs of squeezing at the special angles, at which the cavity lengths were insensitive to the squeezing force, the cavities were robustly mounted and preserved at low vibration sensitivity levels. Although these designs were suitable for transportable applications, they might not be robust enough for space applications. For space application, in 2012, Argence et al. introduced a complex mounting structure to decouple the cavity from the its mounting structure to guarantee a low vibration sensitivity and high robustness at the same time [29]. In 2014, Chen et al. [30] introduced a vibration-insensitive design with a robust and simple mounting structure and might be suitable for space applications. In their design, the cavity was fixed tightly

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in three orthogonal directions using three sets of supports. However, there was not an impact or shake test applied on their design to demonstrate the robustness of the design.

In this paper, we present a new version of Chen's design and demonstrated that the supporting structure held impact accelerations of higher than 100g, with an aluminum cavity model and a ULE cavity spacer.

2 Design of the optical cavity

The schematic of the optical cavity with its supporting structure is shown in Fig. 1. The length of the cavity was 10 cm. The cavity spacer was cut out of a cuboid ultra-low expansion (ULE) glass with a size of $100 \times 60 \times 60 \text{ mm}^3$. Cavity mirrors were made of fused silica to reduce the thermal noise. ULE rings were attached to the rear faces of the mirrors to compensate the zero coefficient-of-thermal-expansion (CTE) temperature shift caused by the fused silica mirrors [31]. The cavity was mounted by squeezing at 12 points on the symmetric planes of the cavity with slim posts. The posts could be divided into three groups. Each group restricted the movement of the cavity in one direction, while had weak influence on the other two. All the supports corporately restricted the movement of the cavity in all directions and kept the cavity a low vibration sensitivity.

The supporting frame of the cavity was made of 7075 Aluminum alloy for reducing the weight. M4 thread holes were made in the frame for mounting the cavity, and screws were used to press the posts to the cavity and kept the cavity fixed. Viton balls were inserted between the screws and the posts and between the posts and the cavity to buffer the force between the cavity and the frame. The squeezing force of the cavity was controlled by the screws. The squeezing of the cavity would introduce a permanent deformation on the

cavity length. The deformation of the frame under acceleration would change the squeezing force on the cavity and, therefore, introduce additional vibration sensitivity to the cavity. To estimate the vibration sensitivity introduced by the frame, we performed finite-element analysis (FEA) simulations to check the influence of the supporting areas displacements to the cavity length and the deformation of the frame under gravity. The simulations showed that, by inserting of Viton balls to the two ends of the posts, the influence of each support area's displacement to the cavity-length change was at the level of $< 0.5 \text{ pm}/\mu\text{m}$ and the deformation of the frame under gravity was at the level $0.1 \text{ }\mu\text{m}/g$ maximum. This result suggested that the vibration sensitivity introduced by the deformation of the frame was at the level of $\leq 10^{-12}/g$.

Because the cavity was supported on the symmetric planes, the cavity-length change under the acceleration was canceled up to first order. A FEA simulation result on the displacement of the cavity along the cavity axis under 1g's cross-cavity acceleration is shown in Fig. 2. It clearly shows that the displacement of the cavity mirror is less than 3 pm, i.e., $< 6 \times 10^{-11}/g$, which is smaller than our experimental result.

The second order of the cavity-length change was introduced by the tilt of the mirrors under cross-cavity acceleration, which could be minimized by adjusting support position along the cavity axis. The relation between the tilt of the mirror and the positions of the support is shown in Fig. 3. The figure shows that cavity mirror had a tilt of less than 0.1 nrad when the support position was at 34.7 mm from center plane. At this support position, the cavity-length change caused by the tilt of the mirror was negligible.

The cavity with the frame was fixed in an aluminum box inside a vacuum chamber. The aluminum box was

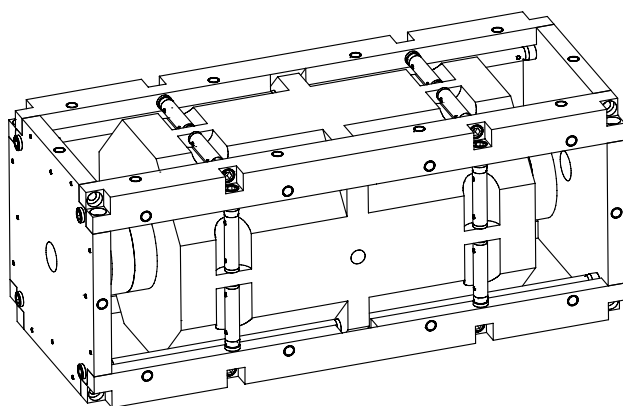


Fig. 1 Schematic of the cavity and its supports. The cavity was mounted by 24 slim posts pressing on the center planes in three orthogonal directions

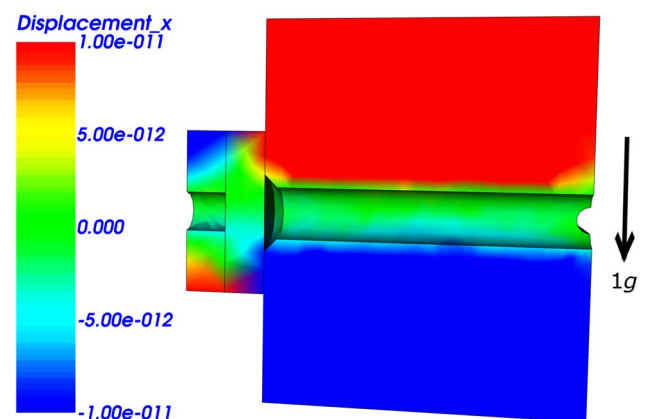


Fig. 2 FEA simulation result of the cavity displacement along cavity axis under 1g's acceleration across the cavity. In this simulation, the position of the horizontal support was 35 mm from center. The unit of the color legend is meter

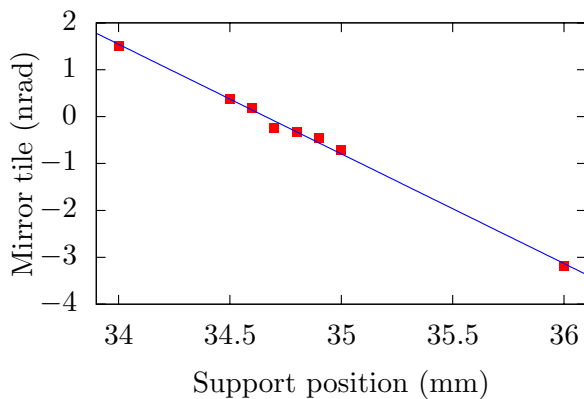


Fig. 3 Relation between the support positions and the tilt of the cavity mirror according to FEA simulation

temperature controlled using thermoelectric coolers (TEC) to guarantee the cavity working at the zero CTE temperature.

3 Robustness test of the mounting structure

An experiment was carried out to demonstrate the robustness of the mounting structure. An aluminum cavity model, which has the same shape and weight with the real cavity, was mounted to the cavity's supporting frame. After attaching tapes with 1 mm's thickness to the edges and corners of the frame, we dropped it from a height of about 250 mm onto an optical table. The model was dropped four times horizontally and one time vertically. A snapshot of the video is shown in Fig. 4. After the droppings, we inspected the

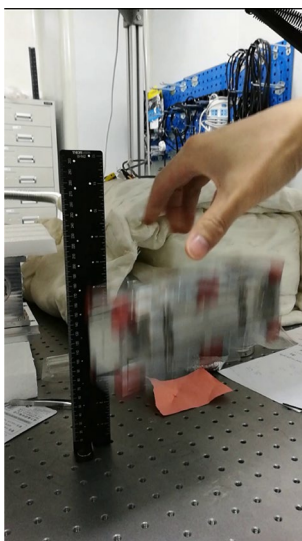


Fig. 4 A snapshot of the video on dropping of the aluminum cavity model mounted in the supporting frame

cavity model and did not observe obvious displacement on the model.

Two methods were used to estimate the impact strength when the frame hit the optical table. First, theoretically, because the thickness of the tape was 1 mm, if we assumed the frame was decelerated uniformly by the tape for 1 mm, the upward acceleration the frame encountered when it hit the table would be $(H/T)g \approx 250g$, in which H was the height from where the frame was dropped and T was the thickness of the tape. This estimation gave the lower limit of the acceleration the frame encountered when hit the table. Second, experimentally, an accelerometer with a range of 2300g was dropped at the same height. The detected accelerations were recorded by an oscilloscope. 100 times repeated dropping were done and the distribution of the impact is shown in Fig. 5. The figure shows that the distribution of the accelerations distributed in a range of 400g to 2300g, and more than 50% of the dropping encountered accelerations of higher than 2000g.

In this test, instead of a ULE cavity, an aluminum model was used. Because ULE glass is very fragile and might be broken under the shock, this test only gave the information that the structure of the supporting frame was very robust. To test the ability of protection to the cavity that the supporting frame supplies, we did a similar dropping test with a ULE cavity spacer mounted in the frame. According to the previous test, the acceleration would be higher than 1000g when dropped from 250 mm. Therefore, in this test, we reduced the dropping height to about 20 mm. The cavity was dropped more than ten times from three directions, and no damage to the ULE spacer was observed after the droppings. Figure 6 is a video snapshot of dropping the cavity. The same accelerometer was used to test the shock acceleration. The statistic of the accelerations that the accelerometer

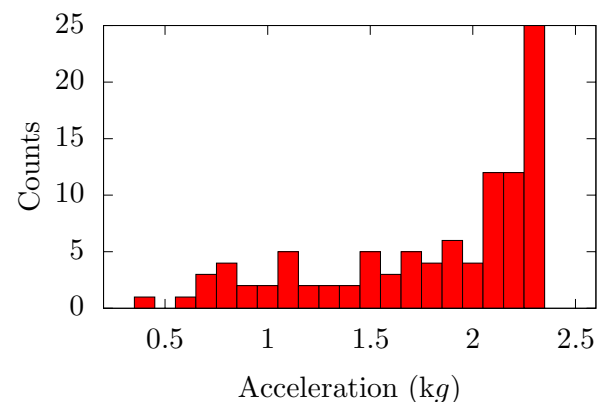


Fig. 5 Distribution of the accelerations that the accelerometer encountered when it was dropped to the optical table at the height of about 250 mm. 2 mm's tapes were attached to the bottom of the sensor to protect the sensor. The high bar at 2.3 kg was caused by the saturation of the sensor

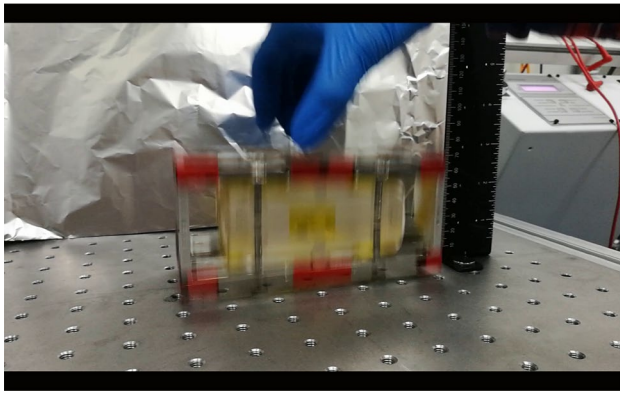


Fig. 6 A video snapshot of the cavity drop from about 20 mm to the optical table

encountered when dropped from 20 mm is shown in Fig. 7, which showed that the accelerations were higher than 200g and with an average of about 650g. Therefore, we concluded that, with this design, the ULE cavity could hold the impacts of higher than 100g when mounted in its frame [32].

We also performed a shaking test on the model mounted in the vacuum chamber with a uniform-distribution shaking acceleration of 6g in a frequency range of 20 to 2000 Hz for 10 min on a shaking platform. There is no displacement of the cavity or damage observed during the test.

4 Characterization of cavity

4.1 Optical property of the cavity

The mirrors of the cavity were plane and concave. The radius of curvature of the concave mirror was 1 m. The mirrors were high-reflectively coated for 1064 nm. The linewidth of

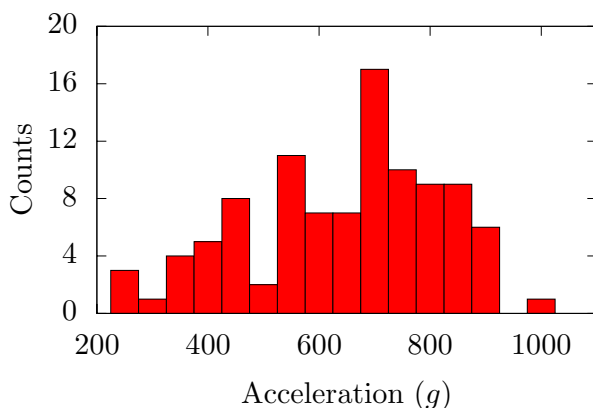


Fig. 7 Distribution of the accelerations that the accelerometer encountered when it was dropped to the optical table at a height of about 20 mm. 1 mm's tapes were attached to the bottom of the sensor

the cavity resonance and the finesse of the cavity were characterized as follows. First, the laser frequency was locked to the cavity resonance. Then, the laser was shut off by using an acousto-optic modulator. The switching speed of the laser was at μs level, which was fast enough for detecting the decay of the laser in the cavity. The photodetector, which detected the reflection from the cavity and was used for laser frequency locking, detected the leakage from the cavity. The decay trace of the leakage was recorded by an oscilloscope, which is shown in Fig. 8. By fitting the leakage trace with an exponential-decay curve, the decay time of the cavity was calculated to be 46.3 μs . The corresponding full width half maximum (FWHM) of the cavity resonance and finesse were calculated to be about 3.4 kHz and 440,000, respectively.

4.2 Vibration sensitivity of the cavity

The vibration sensitivity of the cavity was derived from the relation between the laser frequency variation and the acceleration the cavity encountered during shakings. The measuring procedures were as follows. Another ultra-stable laser was used as a reference laser. An optical setup was built to measure the beating between them. The beating signal was detected by a fast photodetector. Then, the beating frequency, i.e., the frequency difference between them was measured by a frequency counter. At the same time, an accelerometer was fixed to the optical breadboard, on which the optical setup was built, to measure the acceleration. The optical breadboard was laid on an active-vibration-isolation system, which was supported by springs and could be easily shaken.

The breadboard was shaken in three orthogonal directions. The laser frequency difference and the acceleration recorded during the shakings are shown in Fig. 9. From the experiment, the vibration sensitivity of the cavity was derived to be, along the cavity axis (horizontal),

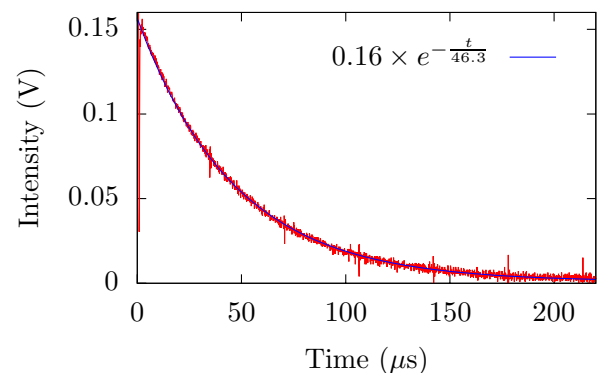


Fig. 8 Leakage of the laser from the cavity obtained by switching the laser off during the laser frequency was locked to the resonance of the cavity

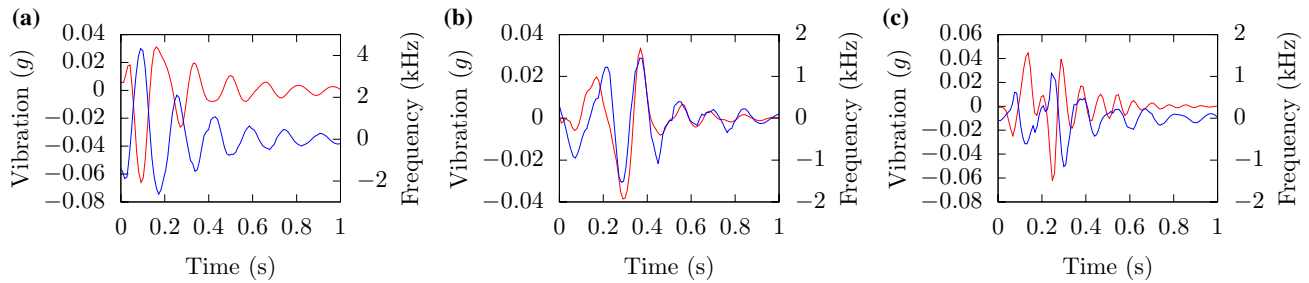


Fig. 9 Laser frequency variation and the corresponding acceleration during the shakes. The figures show the result of the shakes in three directions: **a** horizontal, along the cavity axis. **b** Horizontal, cross the

cavity axis. **c** Vertical, cross the cavity axis. In the figures, the red lines show the vibration accelerations, while the blue lines show the frequency variations

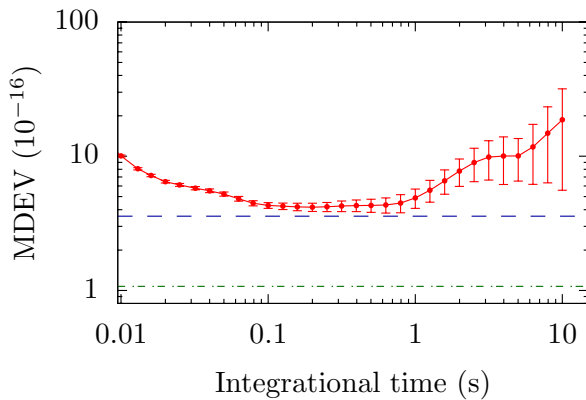


Fig. 10 Modified Allan deviation of the beat note frequency between the two lasers. The dash line and dash-dot line in the figure indicate the thermal noise limit of the 10 cm and 30 cm cavity, respectively

$2.5 \times 10^{-10}/g$; cross the cavity (horizontal), $1.5 \times 10^{-10}/g$; cross the cavity (vertical), $0.8 \times 10^{-10}/g$. The acceleration sensitivities were at the same level with normal laboratory ultra-stable cavities but with a much more robust mounting structure.

The modified Allan deviation of the beat note frequency between the laser, whose frequency was stabilized to this cavity, and a reference laser, whose frequency was stabilized to a 30 cm long cavity, is shown in Fig. 10. The minimal of the Allan deviation is about 4.2×10^{-16} , which is closed to the estimated thermal noise limit of the 10 cm long cavity, as indicated by the dashed line in Fig. 10.

5 Conclusion

In conclusion, we report a new robust and vibration-insensitive optical cavity design. The vibration sensitivities of the cavity were similar to normal vibration-insensitive

optical cavity, while we experimentally demonstrated that the cavity design held impact of higher than 100g. This cavity design was suitable for transportable applications and also a good candidate for space applications.

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